

GIULIA VILLA

MSc Human-Centred Artificial Intelligence · Technical University of Denmark (DTU)

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PROFILE

MSc student in Human-Centred AI at DTU with a BSc in Artificial Intelligence and hands-on experience building and evaluating machine-learning and agentic AI systems. I am interested in turning technology into practical solutions to real human and organisational challenges, understanding the problem first, building thoughtfully, and questioning whether a solution actually works in practice. My current studies combine machine learning and deep learning with applied AI, UX, prototyping, entrepreneurship and sustainability. I moved from Italy to Denmark deliberately to build the next part of my career here. I am curious, active and people-oriented, and I learn best by trying new things, meeting different people and exploring ideas.

EDUCATION

MSc Human-Centred Artificial Intelligence | Sep 2026 – Jun 2028 (expected)

Technical University of Denmark (DTU): Machine Learning, Deep Learning, Applied AI for Entrepreneurs, Developing an Entrepreneurial Mindset, UX & Prototyping, Quantitative Methods to Assess Sustainability. Specialisation: Machine Learning at Scale.

MSc coursework, Artificial Intelligence and Systems | Dec 2025 – Aug 2026

University of Trento: Computer Vision, Image Processing, AI Law & Ethics. Credits recognised by DTU.

BSc Artificial Intelligence — 106/110 | Oct 2022 – Dec 2025

Universities of Pavia, Milano-Bicocca and Milan: Machine Learning, Deep Learning, Data Mining, NLP, Probability & Statistics, Programming, Algorithms & Data Structures. Thesis: deterministic diffusion models for image restoration.

SELECTED PROJECTS

Shipping Exception Agent · DTU · *Applied AI for Entrepreneurs* · 2026

Designed an agentic workflow for freight-forwarding exceptions using Fleet/LangChain, Slack tools via MCP and a human approval gate. It prioritises customer-impacting cases, drafts communications and proposes ranked resolution options, while keeping cost, SLA and customer-contact actions under human control. Focus: reducing coordinator attention load through practical automation.

Cold Diffusion for Image Restoration · *BSc thesis* · 2025

Reimplemented a deterministic diffusion framework on CelebA-HQ with a U-Net to learn the inverse of a non-stochastic degradation process. Achieved PSNR 38.01 and SSIM 0.986 on desaturation, above classical baselines. PyTorch, U-Net · github.com/giuliavillaa/cold-diffusion-restoration

Innovation Minisprint, Winning Team · DTU Skylab & Nordic Women's Health Hub · 2026

Collaborated in a four-person team to develop and pitch AUT, a home urine-testing concept designed to attach to the toilet and share results with GPs through existing healthcare platforms. Won the DTU Minisprint for its approach to accessible testing and reaching at-risk women.

BetWise, Explainable Risk Classification · *Team of 3*

Evaluated 60 configurations across sampling, dimensionality reduction and five classifier families using nested cross-validation. Achieved F1 0.93 with XGBoost, then added SHAP explanations and fairness audits across age and gender. XGBoost, SHAP, scikit-learn · github.com/giuliavillaa/betwise-gambling-risk

EXPERIENCE

Private Tutor: Python & C++ | Milan | 2024–2026

Taught programming from beginner to university level, often debugging alongside students rather than solving problems for them. Developed a practical ability to adapt explanations, communicate and help people become confident solving problems themselves.

TECHNICAL SKILLS

Machine learning: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, diffusion models, U-Net, computer vision, NLP

AI evaluation: Nested cross-validation, leakage detection, class imbalance & SMOTE, fairness auditing, interpretability & SHAP

Agentic AI: Fleet/LangChain, MCP, Slack tools, human-in-the-loop design, instruction-based agent behaviour

Programming & data: NumPy, SciPy, pandas, OpenCV, SQL, C++, R, MATLAB, Git, unit testing, virtual environments, technical documentation

LANGUAGES & OUTSIDE UNIVERSITY

Italian native · **English** C1 (IELTS 7.0) · **Danish** beginner · **German & French** A2

Outside university, I enjoy sport, being outdoors, spending time in nature and meeting new people.

Having moved from Italy to Denmark by choice, I am now building the next part of my life and career here.

Availability: 15–20 hours/week · **Copenhagen / Greater Copenhagen** · **available now** · **EU citizen** · **CPR in place**